

Jack Bajcz

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Education

Michigan State University, MS in Mechanical Engineering – East Lansing, MI Jan 2025 – present

- Tentative Thesis Title: Performance Characterization of a Sub-ppmv Level Freeze-out Purification System for Helium

Michigan State University, BS in Mechanical Engineering – East Lansing, MI Aug 2020 – Dec 2024

- Degree Concentration in Computational Design
- Minors: Computer Science and German

Experience

Graduate Research Assistant, Facility for Rare Isotope Beams – East Lansing, MI Feb 2023 – present

- Engineered complex systems, including CAD assemblies and fabrication drawings for vacuum-jacketed cryogenic transfer lines using SolidWorks, ensuring compliance with ASME B31.3 piping codes.
- Drafted Process and Instrumentation Diagrams (P&IDs) for cryogenic systems.
- Developed internal tools to reduce engineering time spent on repetitive tasks by 30% using VBA.

Engineering and 3D Printing Intern, The Hatch – East Lansing, MI Mar 2023 – May 2024

- Managed a fleet of 3D printers, implementing hardware upgrades that reduced downtime by 50%.
- Provided engineering consultation and support for 10+ student projects

Drafting Intern, Facilities & Operations, University of Michigan – Ann Arbor, MI July 2022 – Aug 2022

- Updated detailed Building Information Models (BIMs) of existing systems using AutoCAD.
- Completed laboratory and roof inspections of 30 campus research facilities, ensuring safety requirements.

Project Support Engineer, Air Force Institute of Technology – Wright-Patterson AFB, OH May 2021 – Aug 2021

- Programmed data collection and analysis tools in Matlab and Python, accelerating academic research.
- Successfully completed extensive background check and was granted secret clearance.

Projects

Heat Exchanger Surrogate Model – ASME IDETC/CIE Student Hackathon Aug 2025

A machine learning model to predict heat exchanger performance metrics without numerical simulation.

- Written in Python, using Tensorflow and Keras.
- 3rd Place Overall in category
- Optimized Solution for consumer-level hardware to augment dataset and run inference.

Liquid Cooling System Steady State Design Tool. – Eaton Aerospace Sept 2024 – Dec 2024

Degree Capstone project to design a steady state analysis tool for liquid cooling systems in aircraft applications.

- Developed a highly extensible MATLAB code library of components and a system-level solver.
- Successfully presented design to Eaton Aerospace engineers and faculty advisors.

Skills

Analysis & Modeling: Large-Scale Cryogenic Systems, Heat Exchangers, Fluid Dynamics, Thermal Systems

CAD & Software: Solidworks, Autodesk Inventor, ANSYS Mechanical, ANSYS Fluent, nTop, Microsoft Office

Programming: Python, MATLAB, LaTeX, VBA, C++

Certificates and Certifications

Engineer In Training (EIT), New York State, 2026

FE - Fundamentals of Engineering, NCEES, 2025

Accelerator Science and Engineering Traineeship in Large-Scale Cryogenic Systems, FRIB/MSU, *In Progress*

Graduate Certificate in Computational Modeling, MSU, *In Progress*

Certificate in Drawing Interpretation (GD&T), ASME, 2026

Certificate in Additive Manufacturing (LPBF), ASME, 2026

Professional Organizations

Cryogenic Society of America (CSA)

American Society of Mechanical Engineers (ASME)

Institute of Electrical and Electronics Engineers (IEEE)

National Society of Professional Engineers (NSPE)

Society of Manufacturing Engineers (SME)